Worksheet 1.4 — The Binomial Theorem

Mathematics: Analysis & Approaches SL (IB) · Unit 1: Number and Algebra

The Big Ideas

The Binomial Theorem expands $(a+b)^n$ without multiplying it out term by term.

- $(a+b)^n = \sum_{r=0}^{n} \binom{n}{r} a^{n-r} b^r$.
- General term: $T_{r+1} = \binom{n}{r} a^{n-r} b^r$.
- $\binom{n}{r} = \dfrac{n!}{r!(n-r)!}$ — Pascal's triangle row n.

Worked Examples

Example 1 — Full expansion

Expand $(x+2)^4$ fully.

Solution. $\sum_{r=0}^{4} \binom{4}{r} x^{4-r} 2^r = x^4 + 4x^3(2) + 6x^2(4) + 4x(8) + 16$.

Conclusion: $x^4 + 8x^3 + 24x^2 + 32x + 16$.

Example 2 — One coefficient

Find the coefficient of x^5 in the expansion of $(x+3)^8$.

Solution. $T_{r+1} = \binom{8}{r} x^{8-r} 3^r$; need $8-r=5 \Rightarrow r=3$. $\binom{8}{3} 3^3 = 56(27)$.

Conclusion: coefficient $= 1512$.

Example 3 — Constant term

Find the constant term in the expansion of $\left(x^2 + \dfrac{2}{x}\right)^6$.

Solution. $T_{r+1} = \binom{6}{r} (x^2)^{6-r} \left(\dfrac{2}{x}\right)^r = \binom{6}{r} 2^r x^{12-3r}$; need $12-3r=0 \Rightarrow r=4$. $\binom{6}{4} 2^4 = 15(16)$.

Conclusion: constant term $= 240$.

Example 4 — Approximation

Use the first three terms of the binomial expansion of $(1+0.01)^5$ to approximate 1.01^5 to 4 decimal places.

Solution. $(1+0.01)^5 \approx 1 + 5(0.01) + 10(0.01)^2 = 1 + 0.05 + 0.001$.

Conclusion: $1.01^5 \approx 1.0510$.

Example 5 — Sum of coefficients

Find the sum of all coefficients in the expansion of $(2x-1)^5$ (hint: substitute $x=1$).

Solution. Substituting $x=1$ gives the sum of all coefficients directly: $(2(1)-1)^5 = 1^5$.

Conclusion: sum of coefficients $= 1$.

Example 6 — Find n from a given coefficient

The coefficient of x^2 in the expansion of $(1+x)^n$ is 28. Find n.

Solution. Coefficient of x^2 is $\binom{n}{2} = \dfrac{n(n-1)}{2} = 28 \Rightarrow n(n-1) = 56 \Rightarrow n^2 - n - 56 = 0 \Rightarrow (n-8)(n+7) = 0$.

Conclusion: $n = 8$ (rejecting the negative root).

Example 7 — Two-variable coefficient

Find the coefficient of x^4y^3 in the expansion of $(x+2y)^7$.

Solution. $T_{r+1} = \binom{7}{r}x^{7-r}(2y)^r$; need $r=3$: $\binom{7}{3}2^3 = 35(8)$.

Conclusion: coefficient $= 280$.

Example 8 — Middle term of an expansion

Find the middle term in the expansion of $(x+3)^6$ (the term where $r=3$, since there are 7 terms total).

Solution. $T_4 = \binom{6}{3}x^3 3^3 = 20x^3(27)$.

Conclusion: middle term $= 540x^3$.

Example 9 — Approximation with a negative term

Use the first three terms of $(1-0.01)^7$ to approximate 0.99^7 to 4 decimal places.

Solution. $(1-0.01)^7 \approx 1 - 7(0.01) + 21(0.01)^2 = 1 - 0.07 + 0.0021$.

Conclusion: $0.99^7 \approx 0.9321$.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Expand $(x+1)^5$ fully.

Question 2

Expand $(2x-1)^4$ fully.

Question 3

Find the coefficient of x^4 in the expansion of $(x+2)^7$.

Question 4

Find the coefficient of x^3 in the expansion of $(3x-1)^6$.

Question 5

Find the constant term in the expansion of $\left(x+\frac{1}{x}\right)^8$.

Question 6

Find the term containing x^6 in the expansion of $(2x-3)^9$.

Question 7

Use the first three binomial terms of $(1-0.02)^6$ to approximate 0.98^6 to 4 decimal places.

Question 8

Find the sum of all coefficients in the expansion of $(3x-2)^4$ (substitute $x=1$).

Question 9

Find the coefficient of x^5y^2 in the expansion of $(x+y)^7$.

Question 10

Find the middle term in the expansion of $(2x-1)^6$.

Question 11 — Challenge

Find the coefficient of x^3y^4 in the expansion of $(2x+y)^7$.

Question 12 — Challenge

Find the value of n if the coefficient of x^2 in the expansion of $(1+x)^n$ is 45.

Question 13 — Challenge

Use the first three binomial terms of $(1+0.005)^{10}$ to approximate 1.005^{10} to 5 decimal places.

Question Recap & Answer Key

Question Recap

1. Expand $(x+1)^5$ fully.
2. Expand $(2x-1)^4$ fully.
3. Find the coefficient of x^4 in the expansion of $(x+2)^7$.
4. Find the coefficient of x^3 in the expansion of $(3x-1)^6$.
5. Find the constant term in the expansion of $\left(x+\frac{1}{x}\right)^8$.
6. Find the term containing x^6 in the expansion of $(2x-3)^9$.
7. Use the first three binomial terms of $(1-0.02)^6$ to approximate 0.98^6 to 4 decimal places.
8. Find the sum of all coefficients in the expansion of $(3x-2)^4$ (substitute $x=1$).
9. Find the coefficient of x^5y^2 in the expansion of $(x+y)^7$.
10. Find the middle term in the expansion of $(2x-1)^6$.
11. Find the coefficient of x^3y^4 in the expansion of $(2x+y)^7$.
12. Find the value of n if the coefficient of x^2 in the expansion of $(1+x)^n$ is 45.
13. Use the first three binomial terms of $(1+0.005)^{10}$ to approximate 1.005^{10} to 5 decimal places.

✓ Answer Key

1. $x^5+5x^4+10x^3+10x^2+5x+1$
2. $16x^4-32x^3+24x^2-8x+1$
3. $\binom{7}{3}2^3=280$
4. $\binom{6}{3}3^3(-1)^3=-540$
5. $T_{r+1}=\binom{8}{r}x^{8-2r}$; $8-2r=0\Rightarrow r=4$: $\binom{8}{4}=70$
6. $T_{r+1}=\binom{9}{r}(2x)^{9-r}(-3)^r$; $9-r=6\Rightarrow r=3$: $\binom{9}{3}2^6(-3)^3=84(64)(-27)=-145152$, so $-145152x^6$
7. $1-6(0.02)+15(0.02)^2=1-0.12+0.006=0.8860$
8. substitute $x=1$: $(3-2)^4=1$
9. $T_{r+1}=\binom{7}{r}x^{7-r}y^r$; need $r=2$: $\binom{7}{2}=21$
10. 7 terms total, middle is T_4 ($r=3$): $\binom{6}{3}(2x)^3(-1)^3=20(8x^3)(-1)=-160x^3$
11. $T_{r+1}=\binom{7}{r}(2x)^{7-r}y^r$; need $r=4$: $\binom{7}{4}2^3=35(8)=280$
12. $\binom{n}{2}=45\Rightarrow\frac{n(n-1)}{2}=45\Rightarrow n^2-n-90=0\Rightarrow(n-10)(n+9)=0\Rightarrow n=10$
13. $1+10(0.005)+45(0.005)^2=1+0.05+0.001125\approx1.05113$